

Fig. 8. A 9V PP3 battery is used to power the circuit. 9V is converted into 5V using a 7805 regulator. The glowing of LED1 indicates the presence of 5V supply in the circuit. The circuit acquires analogue data from the eight channels through IN0 through IN7 inputs. The analogue temperature data at IN0 channel is acquired from LM335 temperature sensor (IC4). Temperature calibration for IC4 is done using a 10-kilo-ohm preset (VR2). The remaining inputs can be fed from external temperature sources.

Assemble the temperature sensor along with preset separately on a small general-purpose PCB for each channel. Extend two wires from each of the general-purpose PCBs to the respective input points (IN1 through IN7) in the main PCB. Two-pin SIL male and female pair connector may

be used for connecting the PCB to the general-purpose PCB for each channel input. As shown in Fig. 5, a 10-kilo-ohm preset is used for calibration of each temperature sensor.

Calibrate each temperature sensor (LM335) before connecting the circuit to the PC. After calibration is done, install the sensors at appropriate locations or on the device whose temperature is to be monitored. (All the relevant files of the GUI software are stored in 'EFYproject/VB Project' folder of 'D' drive.) Now, run the datalogger8chnl.exe GUI software and click 'start' button to start the data acquisition and logging process. If data display on the dashboard is not proper, press reset switch S1 momentarily, or switch off the power supply and then switch it on. Using preset VR1, adjust ADC reference voltage such that it is

exactly 5V across pin 32 of IC3.

Microcontroller firmware

The main.c source code for ATmega32 (given at the end of this article) is written in 'C.' It is compiled using avr-gcc cross-compiler to generate hex code. Avrdude is used to burn hex code into the ATmega32 microcontroller.

WinAVR is a free software development tool for AVR series microcontrollers hosted on the Windows platform. It has avr-gcc, avr-libc, avr-binutils and avr-dude within one package. Linux users have to install these components separately. For details of AVR programming hardware and software, visit 'www.electronics.com'

EFY note. All the relevant source codes of this project have been included in this month's EFY-CD.

MAIN.C

```
#include<avr/io.h>
#include<util/delay.h>

void UART_transmit(unsigned char data)
{
    while(!((UCSRA & (1<<UDRE))));
    UDR=data;
}

void adc_getdata(void)
{
    unsigned int temp;
    unsigned char ch=0;
    while(1)
    {
        for(ch=0;ch<8;ch++)
        {
            AD_CONVERTER=0x00+ch; // Selecting Chan-
```

```
nel
    ADCSRA=0xE7; // ADC free running
    mode, 128 prescaling division factor
    temp=(ADC)*(48);
    temp=temp/10;
    temp=temp-2280;
    temp=temp/10;
    _delay_us(1000);
    UART_transmit(ch*0'); // Sending
    channel number for synchronization
    _delay_ms(10);
    UART_transmit(temp); // Sending
    temperature value
    _delay_ms(1000);
    }
}
```

```
int main(void)
{
    DDRA=0x00;
    PORTA=0x00;

    UCSRA=0;
    UCSRB=1<<TXEN; // UART transmit enable
    UCSRC=1<<URSEL | 1<<UCS21 | 1<<UCS20;
    // 8 data bit, a stop, none parity
    UBRRH=0;
    UBRRL=103; // for 9600 baud at 16MHz

    adc_getdata();

    return 0;
}
```

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